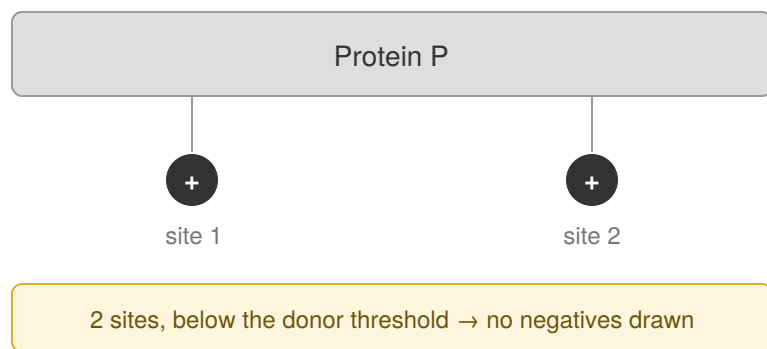


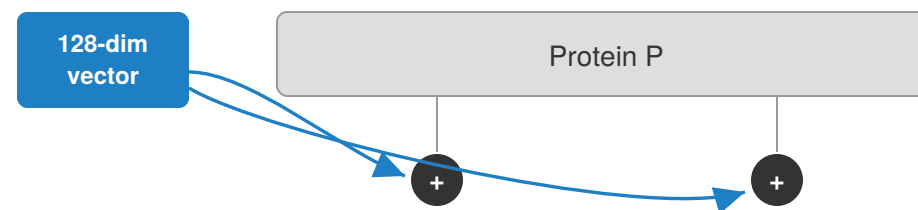
(a) Sampling threshold creates label-homogeneous proteins



Every site of P carries the same label

P is label-homogeneous

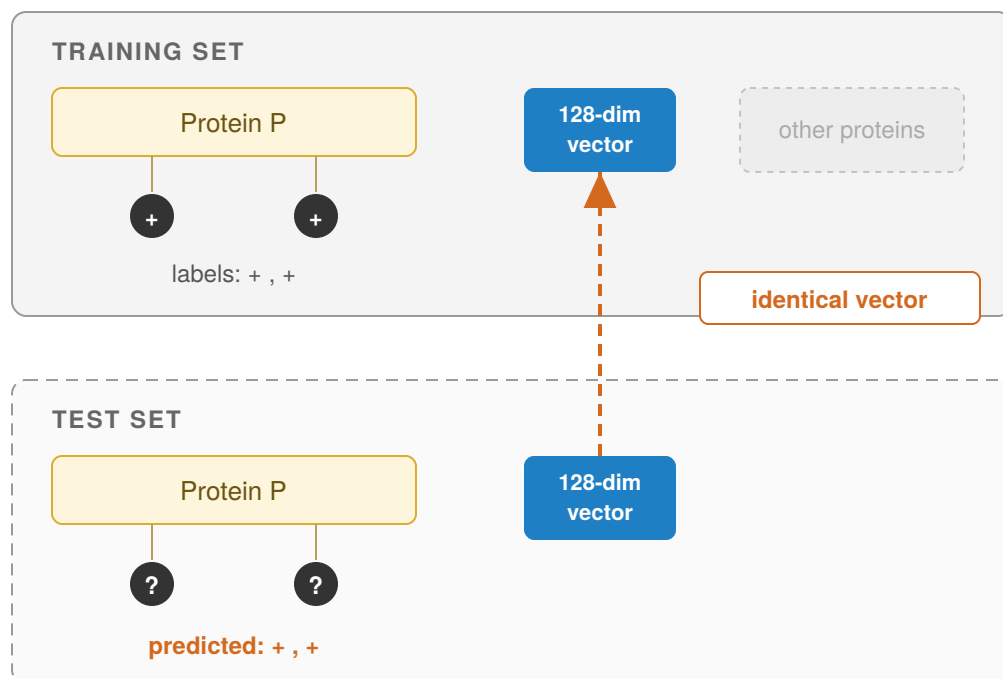
(b) A protein-level feature is constant within P



the same vector is attached to every candidate site

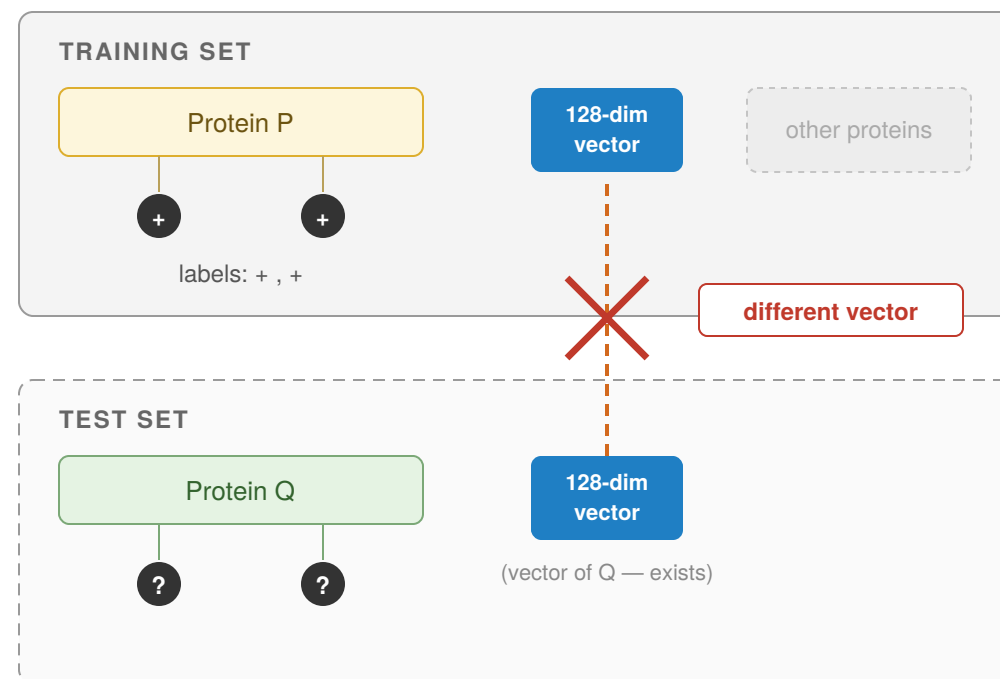
it distinguishes proteins, never sites within a protein

(c) Site-level split — the vector becomes a key



P appears in both partitions → the vector recalls its labels

(d) Protein-disjoint split — the key opens nothing



Q never appears in training → its vector recalls no label